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Career Objective

Python Engineer building production-grade AI agents and LLM applications with LangGraph, RAG, and MCP. Hands-on experience **shipping containerized AI services to AWS** – Docker, HTTPS reverse proxies, observability. Won PhonePe's AI Immersion Hackathon. Backed by production experience – automation pipelines on HDFS, on-call across 6 microservices, and FastAPI backends with PostgreSQL.

Education

Bachelor's Amrita Vishwa Vidyapeetham University, Coimbatore, India
• B.Tech in Computer Science and Engineering (CGPA: 7.29)

08/2020 – 05/2024

Skills

AI/GenAI: LangChain, LangGraph, RAG, FAISS, Agentic AI, MCP (Model Context Protocol), Sentence Transformers, Cross-Encoder Reranking, Prompt Engineering, LangFuse (observability), LLM APIs (Groq, OpenAI)

Languages: Python, SQL, Shell Scripting

Backend: FastAPI, REST API Design, Async Python, PostgreSQL, Redis, Pydantic, JWT Authentication, Rate Limiting

Cloud & DevOps: AWS EC2, Docker (multi-stage builds), Caddy reverse proxy, Let's Encrypt, Linux/Ubuntu, UV (modern Python packaging), Ruff, PyRight, Make

Data & Infra: PySpark, Apache Hive, HDFS, Airflow, Cron, Git, Postman

Work Experience

PhonePe [🔗](#), Product Solution Engineer

India's leading digital payments platform – Quick-Commerce division (Pincode)

Bengaluru

07/2025 – Present

- **Won the PhonePe "AI Immersion Hackathon"** – built a complete end-to-end appointment booking system (merchant-facing: salons, spas, etc.) in 5 days using AI-assisted development. Led architecture design, API contract definition, and backend integration; judged on production-readiness and code quality.
- Built an **end-to-end automation pipeline** (Python + PySpark + Hive) replacing a paid third-party platform – extracts merchant data from HDFS, applies business logic, and creates tasks via internal REST APIs. Runs **twice daily via cron** with 99%+ uptime and zero manual intervention.
- Provided **L1/L2 on-call support across 6 distributed microservices** (Catalogue, Transaction, Fulfillment, Search, Buyer/Seller Apps). Achieved best-in-team metrics: **MTTA < 1hr, MTTR < 13hrs** through systematic log analysis and cross-service debugging.
- Served as **primary technical POC for merchant API integrations** (Order, Inventory, Catalog APIs) – led end-to-end lifecycle from consultation to production sign-off with **100% success rate**. Debugged API request/response flows using Postman and log analysis tools.
- Built an **automated reporting pipeline**: Google Sheets API → HTML templating → PDF generation → SMTP email delivery, eliminating daily manual reporting work.

Jobshine [🔗](#), Python Developer | Chatbot Engineer

Singapore-based startup – blue-collar recruitment automation

Singapore (Hybrid)

05/2024 – 05/2025

- Built and maintained **AI chatbots across WhatsApp, Messenger, and Telegram** using Python and Make (Integromat); automated candidate engagement workflows and job application pipelines for messaging platforms.
- Developed **in-house Python web crawlers** (Selenium, BeautifulSoup, Requests) replacing a paid scraping tool, saving **100%** in tool costs and reducing manual data collection by **70%**.
- Performed **EDA on engagement data** using Pandas to improve lead quality and recruitment conversion outcomes.

Projects

Jira Coding Agent – Deployed AI Agent on AWS (04/2026 - Present) [Project link](#) [🔗](#) [Live demo](#) [🔗](#)

- Built an **autonomous AI coding agent** using LangGraph that watches Jira for tickets, modifies a React codebase via RAG (FAISS + grep), runs tests with a **self-healing retry loop**, and creates PRs with **before/after screenshots via Playwright**

MCP. Includes human-in-the-loop approval for high-risk changes and full LLM observability via **LangFuse**.

- **Deployed to production on AWS EC2** (ap-south-1) — multi-stage Docker build, Caddy reverse proxy with Let's Encrypt HTTPS, Elastic IP, custom subdomain. Optimized container from 6 GB to 800 MB (87% reduction) by routing torch to CPU-only PyTorch index.

Bubble – Production-Grade RAG Chatbot (04/2025 - Present) [Project link](#)

- Built an **agentic RAG chatbot** using LangChain, LangGraph, and FAISS with a **two-stage retrieval pipeline**: bi-encoder (FAISS) for fast recall → cross-encoder reranking for precision, with configurable relevance thresholds.
- Designed **config-driven architecture** with Pydantic validation and YAML config – switching LLM providers, embedding models, or retrieval parameters requires zero code changes. Improved chunking from 322 noisy fragments to 82 high-quality chunks.

FinSight – Stock Portfolio Tracker & Sentiment Analyzer (05/2025) [Project link](#)

- Full-stack application with **FastAPI backend** (async endpoints, JWT auth, dependency injection), **PostgreSQL** database (Neon), real-time P&L tracking via yFinance, and news sentiment analysis using FinBERT. Deployed with Streamlit UI.

Certificates

Unsupervised Learning, Recommenders, Reinforcement Learning (06/2025) [DeepLearning.AI](#) | [Coursera](#)

Advanced Learning Algorithms (05/2025) [DeepLearning.AI](#) | [Coursera](#)

GenAI Intensive Course (04/2025) [Google](#) | [Kaggle](#)

Supervised Machine Learning: Regression and Classification (02/2023) [DeepLearning.AI](#) | [Coursera](#)

Publications

A Multi-Round Zero Knowledge Proof Algorithm for Secure IoT and Blockchain Environments [DOI](#)

Published on International Journal of Safety and Security Engineering